

Lesson 10 — Design Challenge — Color Signal Machine

Matches LEGO Teacher Portal lesson: "Surprise Squad".

Objective (I can...): I can design a machine that uses color to stand for a message, and the motor acts on it. **CS concept:** Using color to represent information (a color means something) · **Builds on:** Lessons 6-9. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): Colors carry messages everywhere — red means stop, green means go. Computers use colors and signals to stand for information too. *Ask the group:* "What message could YOUR machine's colors send?"

Materials: LEGO CS & AI kit + device per pair, plan mats, color cards. **Model already built** (Hardware machine from Lesson 6) — the 25 min is plan + code + play, small changes only.

60-minute flow - Warm-up (10): Hardware review — pairs name the two parts (motor, color sensor) and do a quick action for each ("spin!" "I see red!"). (*Theme: Signal fair.*) - **Concept + demo (8):** Challenge: build a "color signal machine" where each color means something and the motor reacts (green = open the gate, red = close it, yellow = wiggle). Coach says a plan for one color, then hands off. - **Build/Code (25):** Pairs plan and code a machine where at least two colors each send a different message to the motor, on the built machine. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a third color-message, or add a light so the signal is easy to see. - **Share + tidy-up (10): Signal fair** — pairs show their color cards and the machine obeying each one. The audience guesses what each color means before the pair runs it. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator works the cards and points to the plan; swap at the wiggle break. (*Trio rotates plan/code/test.*)

Differentiation: *Support* — a two-color target with a partial plan. *Extension / fast-finisher* — three colors with three different motor jobs. *Reteach* — reuse the Lesson 9 program and change what each color does.

Exit check: Pair shows a working machine where colors stand for messages the motor acts on, with a matching plan. **Progress layer:** Send home the **Unit 2 parent note** ("your child knows the motor and color sensor and can make them work together") with the **build photo**. **At-home (optional):** Make up a color code for your room (green = come in, red = knock first); draw it. ~5 min.